Water Quality C - Water Quality - MIT Invitational - 01-22-2021

If at any point, Scilympiad is having trouble saving longer answers (short answer, etc.) please email your response to scioly@mit.edu with the title "[Team #]-[Event Name]-[Question #]" with any other details identifying the question in the body of the email.

Use this table to answer questions 1-2

	Species (Number Present)							
Location	Banded Coral Shrimp	Pencil Urchin	Humphead Wrasse	Triton	Gorgonia	Sponge	Flamingo Tongue Snail	Nassau Grouper
A	340	98	69	839	0	121	0	0
B	290	257	0	31	721	211	86	594

1. (5.00 pts)
Calculate Jaccard's Index for coral reef ecosystems A and B to three decimal places. Show all of your work. Explain what the value represents and how a scientist can use the data within their research of these two locations. (5)

Expected Answer: 1/(1+3+5)=0.625 (one point for each side of the equation. One additional point for three decimal places in the final answer). A scientist can use this data to measure similarities between ecosystems (1). Any additional valid use of this in scientific practice can earn an additional +2 points IF TEAMS DO NOT GET THE CALCULATION, NO CREDIT IS AWARDED

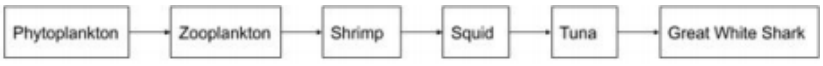
2. (2.00 pts) Explain how each of these ecological computations can be used by a scientist to understand similarities between ecosystems (2)

a. Morisita's Overlap Index

b. Renkonen Index

Expected Answer: Morisita: Measures the dispersion diversity within a population Renkonen: shows the dissimilarity between two locations

The food web below, consisting of every organism in the ecosystem, is used for questions 3-8



3. (1.00 pts)
If phytoplankton receive 456000J of energy from the sun, how much energy would the Great White Shark receive, in kJ? Please leave off the units in your written answer. If the value is less than zero, include a '0' before the decimal place (e.g. write one-tenth as '0.1', not '.1'

0.00456

4. (1.00 pts) Which organism would be considered the primary producer?

Phytoplankton

5. (1.00 pts)
What is the name of the law used to describe the transfer of energy within this ecosystem? Make sure you write your answer using letters if the answer contains numbers

Ten | Percent | Law

6. (2.00 pts) Describe why a Shrimp would assimilate a smaller proportion of energy from the Zooplankton than the Great White Shark would assimilate from the Tuna

Expected Answer: It is easier to digest prey items that are similar to the predator and are close in evolutionary history (1). In turn, the tuna is more closely related to the shark than shrimp is to zooplankton (1).

7. (1.00 pts) An additional organism is introduced into this ecosystem. It is described as a secondary consumer. Which organism will it compete with for the same prey item?

Shrimp

8. (1.00 pts) An additional organism is introduced into this ecosystem. It is described as a secondary consumer. Which organism will it consume?

Zooplankton

A manager of a potable water treatment facility has to balance a variety of factors in order to minimize issues that occur during operations. Match the factor (answer choices) with the problems that arise from them (questions 9-17). **Write the letter of the answer in the box**. Answers can be used once, more than once, or not at all. Each question is worth (1) point.

A. Poor Primary Classification

B. Hydraulic Overload

C. Nitrification

D. Organic Overload

E. Organic Underload

F. Nutrient Shortage

9. (1.00 pts) Lowest pH

C

10. (1.00 pts) Rising sludge

F

11. (1.00 pts) Highest Energy Use

E

12. (1.00 pts) Filamentous bacteria

F

13. (1.00 pts) Poorest effluent quality

D

14. (1.00 pts) Plugging

A

15. (1.00 pts) Highest Chlorine Demand

C

16. (1.00 pts) Lowest DO Levels

D

17. (1.00 pts) Standing Water

A

18. (5.00 pts) Describe the process of Ekman transport and how this phenomena relates to upwelling

Expected Answer: Ekman transport is caused by the Coriolis effect (1) and drives ocean gyre circulation (1). Ekman transport deflects water at a 90 degree angle from the wind (1). In some areas, it pulls water away from shore, causing upwelling (1). In other areas, it draws water in toward shore and sparks downwelling (1)

19. (8.00 pts) Describe the term 'foraging arena' and explain how the concept of an optimal forager relates to the foraging arena

Expected Answer: A foraging arena is an area where vulnerable prey is available and ready to be consumed (1) and is where predators go to search out their prey (1); when lower-level trophic organisms enter this area in search of prey, predators higher up the food chain are also looking for prey items (1). Organisms come out of hiding to enter their foraging arena, leaving behind the reduced threat of being eaten in order to find their next meal (1). An optimal forager takes advantage of the foraging arena (1). These organisms take into account the energy expended to consume a prey item, how much energy a prey item will give them, and how far away from their safe area they wander to find food/(1). The goal of an optimal forager is to maximize the benefit of searching for prey in the foraging arena while minimizing the risk of being eaten by leaving its hiding place(1). Organisms may school or shoal to make themselves less vulnerable(1)

20. (1.00 pts) Select all of the following which are metals with maximum contaminant levels in water as designated by the EPA

(Mark **ALL** correct answers)

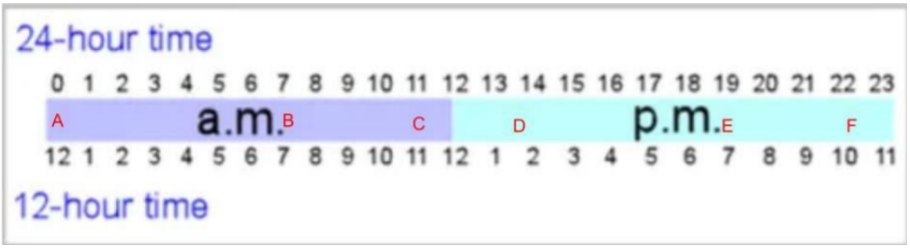
- [x] A) Barium
- [x] B) Cadmium
- [ ] C) Colbalt
- [ ] D) Vanadium

- [ ] E) Copper
- [x] F) Chromium

21. (5.00 pts) Describe the 'Deserts of the Ocean.' Where are these systems located? Describe what processes contribute to the formation of these 'deserts.'

Expected Answer: This location is the center of an oceanic gyre (1). These areas are located in a high-pressure zone, causing water to go away from the area(1). In turn, oceanic downwelling occurs(1). This takes organic carbon and other nutrients down to depth where they cannot be used by phytoplankton(1), making productivity very low (1).

22. (1.00 pts) Select the option(s) in which a crepuscular feeder would most likely feed



(Mark **ALL** correct answers)

- [ ] A) A
- [x] B) B
- [ ] C) C
- [ ] D) D
- [x] E) E
- [ ] F) F

23. (1.00 pts) Select the options where the terms are correctly matched with their definitions

(Mark **ALL** correct answers)

- [ ] A) Thunniform swimming: A type of s-wave swimming where the s-wave starts at the base of the head
- [x] B) Tethys Sea: an equatorial sea, surrounded by the global ocean Panthalassa, that existed during much of the Mesozoic Era to the east of Pangea and then between the ancient continents of Gondwana and Laurasia; region of modern Indo-Pacific
- [x] C) TMAO: a compound found in many elasmobranchs used to buffer kidney protein pH and keep the proteins from denaturing
- [x] D) Aggressive mimicry: when predators closely resemble objects to lure unsuspecting prey close so they can be ambushed by the predator
- [ ] E) Browsing predator: organism that lives as an adult in freshwater but migrates to the ocean to consume prey
- [x] F) Trophic guild: a group of species that exploit the same diet resources or exploit different resources in a similar way

24. (6.00 pts) What is the deep scattering layer? How was this discovered? What ecological phenomena does the deep scattering layer show?

Expected Answer: The deep scattering layer displays the migration of biomass in the water column(1). Vertical migrants(1) rise to the surface during the night(1) and sink to depth during the day (1). This was discovered shortly after WWI (1) when scientists were using radar to attempt to detect German U-Boats (1).

25. (1.00 pts) The EPA creates a Contaminant Candidate List every _____ years. Use a numeric value (e.g. '0' instead of 'zero')

5

For Questions 26-28, identify whether the potable water treatment facility should use sulfate-reducing bacteria (SRB), oxidation ditches, or extended aeration

26. (1.00 pts) Located in an area with harsh weather conditions, able to denitrify waste, and able to continue operations when shock occur

- A) sulfate-reducing bacteria (SRB)
- B) oxidation ditches
- C) extended aeration

27. (1.00 pts) Easy to operate and install, infrequent disposal of sludge, and a small ecological footprint

- A) sulfate-reducing bacteria (SRB)
- B) oxidation ditches
- C) extended aeration

28. (1.00 pts) The owner wants to be able to operate the facility remotely and treat toxic waste streams. She also want to deal with the removal of phosphorus

- A) sulfate-reducing bacteria (SRB)
- B) oxidation ditches
- C) extended aeration

29. (2.00 pts) A high KH indicates a high presence of two polyatomic ions in water. What are they?

Expected Answer: Carbonate and bicarbonate

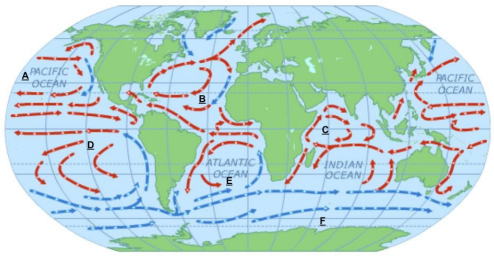
30. (1.00 pts) True or false: Volcanoes, geysers, and hot springs will cause the water in the surrounding area to become more alkaline

- True
- False

31. (1.00 pts) Which of the following options orders the steps of the predation cycle in the correct order?

- A) Search, encounter, detect, attack, digest, eat
- B) Search, encounter, detect, attack, eat, digest
- C) Encounter, search, detect, attack, digest, eat
- D) Encounter, search, detect, attack, eat, digest
- E) Search, detect, encounter, attack, eat, digest
- F) Search, detect, encounter, attack, digest, eat

Use this image for Question 32



32. (1.00 pts) Which location on the map does not have one of the five oceanic garbage patches?

- A) A
- B) B
- C) C
- D) D
- E) E
- F) F

33. (1.00 pts) Most marine biodiversity is found in which area of the water column?

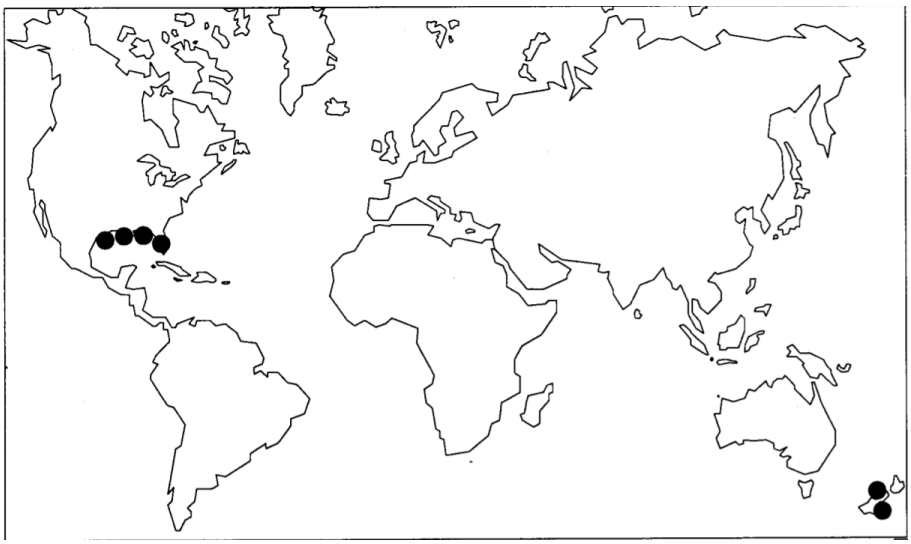
- A) Pelagic
- B) Benthic
- C) Epipelagic
- D) Cold, shallow water
- E) Warm, shallow water

34. (1.00 pts) Which ions are found with concentrations greater than 10^{-3} molarity in seawater?

(Mark **ALL** correct answers)

- [ ] A) Strontium
- [x] B) Sulfate
- [x] C) Magnesium
- [ ] D) Fluoride
- [x] E) Chloride
- [ ] F) Bromide

35. (1.00 pts)
This map shows common occurrences of Karenia breve, a harmful algae. Which type of toxin is produced by this organism and what is the name of the human health effect that goes along with it



- A) Brevetoxin, Diarrhetic Shellfish Poisoning
- B) Brevetoxin, Neurotoxic Shellfish Poisoning
- C) Neuratoxin, Amnesic Shellfish Poisoning
- D) Neuratoxin: Neurotoxic Shellfish Poisoning
- E) Karenitoxin, Amnesic Shellfish Poisoning
- F) Karenitoxin; Paralytic Shellfish Poisoning

36. (1.00 pts) Mangroves are considered a keystone species in many estuarine ecosystems. Which of the following are ecosystem services provided by mangrove forests?

(Mark **ALL** correct answers)

- [x] A) Decrease turbidity
- [x] B) Remove toxins from the water
- [x] C) Raise dissolved oxygen levels in the water
- [x] D) Increase coastline stability
- [x] E) Reduce creep from storm surges and king tides
- [x] F) Serve as a nursery for juvenile fish

37. (2.00 pts) What are the two most likely reasons a septic tank disposal system may fail?

Expected Answer: The tank is not emptied (1), the soil drainage is poor (1)

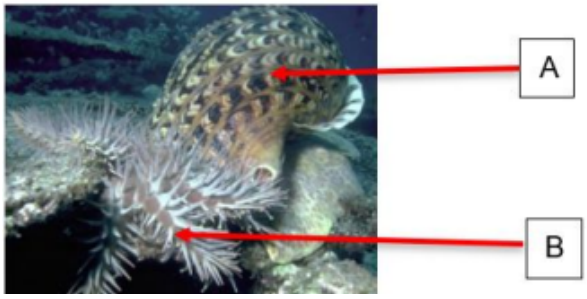
38. (1.00 pts) What is the name of the algae associated with corals?

zooxanthellae

39. (1.00 pts) Which step in potable water treatment is most vulnerable to algal blooms?

Sedimentation

Use this image for questions 41-43



40. (2.00 pts) Identify Organisms A and B

Expected Answer: A: Triton B: Crown-of-Thorns Starfish

41. (2.00 pts) Describe the relationship between organism A and B

Expected Answer: Organism A/Triton is consuming/predating/eating/preying on Organism B/Crown of Thorns Starfish. All or nothing

42. (1.00 pts) Describe how Organism B can negatively impact coral reef ecosystems

Expected Answer: Coral is the main component of its diet

Use this table for questions 43-47

	Species (Number Present)							
Location	Banded Coral Shrimp	Pencil Urchin	Humphead Wrasse	Triton	Gorgonia	Sponge	Flamingo Tongue Snail	Nassau Grouper
A	340	98	69	839	0	121	0	0
B	290	257	0	31	721	211	86	594

43. (1.00 pts)
Based on the table above, is location A in the Pacific, Atlantic, or in another region around the globe? ("Another region" is sufficient; the specific area does not need to be named.)

Pacific

44. (2.00 pts) Justify your answer to the previous question

Expected Answer: The presence of the humphead wrasse

45. (1.00 pts)
Based on the table above, is location B in the Pacific, Atlantic, or in another region around the globe? ("Another region" is sufficient; the specific area does not need to be named.)

Atlantic

46. (2.00 pts) Justify your answer to the previous question

Expected Answer: The presence of Gorgonia, Flamingo Tongue Snail, and Nassau Grouper. +1 for one organism, +1 for the other two

47. (1.00 pts) Describe how El Nino cycles can impact populations of algae, particularly algae in the Pyrodinium genus

Expected Answer: During El Nino years, the cell density of algae tend to increase substantially compared to normal and La Nina conditions

48. (1.00 pts) What are the three sexual reproductive modes of green algae?

Expected Answer: Isogamy, anisogamy, and oogamy (+.5 for one, +.75 for two)

49. (1.00 pts) Select the pigment(s), if any, that are present in high abundance in green algae

(Mark **ALL** correct answers)

- [x] A) Chlorophyll a
- [x] B) Chlorophyll b
- [ ] C) alpha-carotene
- [ ] D) β carotene
- [x] E) Lutein
- [ ] F) Neoxanthin

50. (1.00 pts) Vaucheriaxanthin is unique to which fleshy algae group

(Mark **ALL** correct answers)

- [ ] A) Red
- [ ] B) Yellow
- [ ] C) Brown
- [ ] D) Green

- [x] E) Yellow-Green
- [ ] F) Yellow-Brown

51. (1.00 pts) Describe the role of chrysolaminarin in Xanthophyta algae

Expected Answer: It is the primary storage molecule

52. (2.00 pts) Describe the role of silica in Xanthophyta algae

Expected Answer: Cell walls (1) are filled with it to increase rigidity/cell shape/cell structure (1)

53. (1.00 pts) Which two areas are the center of marine fish biodiversity in the world?

(Mark **ALL** correct answers)

- [ ] A) Mediterranean Sea
- [ ] B) Pacific Northwest in the United States and Canada
- [ ] C) Off the coast of South Africa
- [x] D) The Indo-Pacific
- [x] E) The Caribbean
- [ ] F) Antarctica

54. (6.00 pts) Explain how plate tectonics impacted the movement of aquatic organisms between these two areas from the period of 225mya to today

Expected Answer: The Tethys Sea was the area of greatest biodiversity (1). Until about 2-4mya (1), there was a passage connecting the Carribean and Tethys Sea (1) through the Medditerranean (1). The same is true of the Ismuth of Panama (1). Once these passageways closed, the exchange of species between these areas slowed (1). If they chose the wrong areas, this question will be marked incorrect

The image below displays Organism C



55. (1.00 pts) What is the common name of Organism C, as stated on the rules

Gorgonia

56. (4.00 pts) Describe the feeding of Organism C, as well as place them in a feeding category

Expected Answer: They feed passively (1) on zooplankton (1), particularly at night (1). This makes these organisms carnivores (1)

57. (3.00 pts) Explain the use of gorgonin in this organism

Expected Answer: It is a scleroprotein (1) used for structure (1) to keep the branches upright (1)

58. (1.00 pts) This organism would be protected under the Coral Reef Preserve Act of 1966

○ True ◉ False

59. (1.00 pts) Would you find this organism in the Atlantic, Pacific, or Worldwide

Atlantic

60. (5.00 pts) Describe the Fleshy Macroalgal Index and how ecological conditions can impact the value the index produces, as well as how it is calculated

Expected Answer: The FMI measures large fleshy algae/seaweed on a reef (1) it is calculated by multiplying the area covered by the algae (1) with the height of the algal canopy (1). A higher FMI generally means there are fewer grazers (1) and higher nutrient availability/load (1)

61. (1.00 pts)
What is the common name of the oceanic area that encapsulates the coasts of the Solomon Islands, Malaysia, Indonesia, Papua New Guinea, Timor-Leste, and the Philippines?

Expected Answer: The Coral Triangle

62. (5.00 pts) What are the benefits of a secondary male in a hermaphroditic species? How do they differ from primary males?

Expected Answer: These organisms are larger and older (1) which displays a high survivorship/good genetics (1). Often can directly inseminate a female instead of broadcast spawning (1). A primary male stays as a male throughout their entire lives (1) while a secondary male transitions from a female to a male late in life (1)

63. (3.00 pts) Compare and contrast between a protandrous and protogynous species

Expected Answer: Hermaphroditism (1); Protandrous: Switches from male to female (1); Protogynous: Switches from female to male (1)

64. (1.00 pts) What time period is commonly referred to as the ‘Age of Fishes?'

Devonian

65. (4.00 pts) Differentiate between a ‘resilient’ and ‘resistant’ organism when it pertains to pressure from fisheries

Expected Answer: Resistance relates to the ability of a species to maintain numbers as it is being fished (1) and relates to the volume of eggs produced per year (1). Resilience refers to the ability of a species or population to bounce back after they have been farmed (1) and is related to the speed at which organisms reach sexual maturity and can produce viable offspring (1).

66. (15.00 pts)
Traditionally, studies separate species into r-strategists and k-strategists. However, conventional literature in aquatic sciences use three different categories to place marine species. Name these categories and describe the life history characteristics of these groupings. Using this information, define "resilience" and "resistance", and evaluate whether each category would be considered resilient, resistant, both, or neither to fisheries pressure based on your answer to the previous question. (15)

Expected Answer: Resistance is the ability to maintain numbers as a species is fished (1) and relates to the volume of eggs per year (1). Periodic life strategists are the only resistance group (1), as iteroparous batch spawners (1) have many egg classes per year to spread mortality over (1) . Opportunistic and equilibrium strategists do not have as many reproductive events (1) and are periodic reductive strategists (1), so they are not resistant to fisheries(1) . Resilience is to the ability of a population to recover after they have been farmed(1). It is related to the rate at which organisms reach sexual maturity (1). Opportunistic life strategists are the only resilient group (1), as they fill up their age structure quicker than the other two groups (1) since they mature faster(1). Periodic life history strategists are resistant but not resilient (1), opportunistic species are resilient but not resistant(1), and equilibrium strategists are neither resistant or resilient to fishery pressure (1).

The image below is Organism D



67. (1.00 pts) What is the common name of organism D, as stated on the rules

Grouper

68. (2.00 pts) Would Organism D be considered a neritic species? Why or why not?

Expected Answer: Yes (1); reefs are often associated with the continental shelf and the littoral zone (1)

69. (3.00 pts) Explain the benefit of oscillatory swimming to Organism D

Expected Answer: This organism is an ambush predator (1). Oscillatory swimming allows this organism to stay stationary in the water column (1) and attack unsuspecting prey (1)

70. (1.00 pts) Would Organism D be considered a generalist or a specialist?

Generalist

71. (1.00 pts) Explain the role of seagrass beds to this species

Expected Answer: Juvenile groupers/Organism D use it as protection from predation

The image below showcases Organism E



72. (1.00 pts) What is the common name of Organism E, as stated exactly on the rules?

Barramundi Cod

73. (1.00 pts) This organism is considered a demersal species. Define the term ‘demersal’

Expected Answer: An organism that is associated with, but does not live on, the ocean floor

74. (1.00 pts) Organism E stumbles upon a region with lively vegetation and a seafloor full of clay and sandy sediments. This habitat ideal for this organism

○ True ◉ False

75. (1.00 pts) Once laid, Organism E will guard their eggs until it hatches

○ True ◉ False

76. (1.00 pts) Organism E is considered a part of a monophyletic group. What is a monophyletic group?

Expected Answer: Every organism within the taxonomic grouping of Organism D/Barramundi cod has the same common ancestor

77. (2.00 pts) What is the IUCN? What status was this species given by the IUCN?

Expected Answer: International Union for the Conservation of Nature (1); data deficient (1)

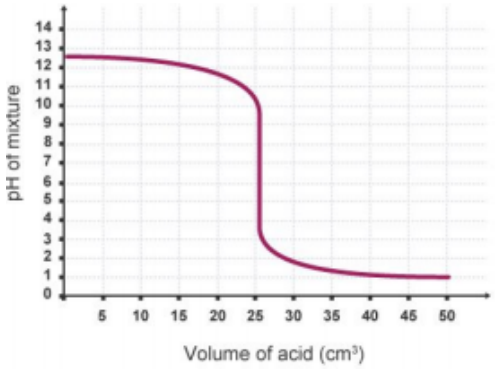
78. (1.00 pts) This organism is in the Class Actinopterygii. What does this show about its evolutionary advancedness?

Expected Answer: It is in the most advanced class of fishes

79. (3.00 pts) Differentiate between a holotype, paratype, and neotype

Expected Answer: Holotype: The first organism of a species discovered in which the species is named from (1); Paratype: Organisms that exhibit similar characteristics as the holotype and are then used in museums and research as equivalents to the holotype (1). Neotype: A specimen used to replace a missing or damaged holotype (1)

Use this image for questions 80-81



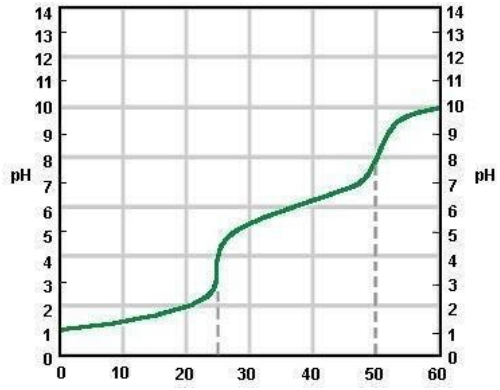
80. (1.00 pts) What is the pH of the equivalence point?

7

81. (1.00 pts) Based on the graph, a _____ is being titrated with a _____?

- (●) A) Strong acid, strong base
- ( ) B) Strong acid, weak base
- ( ) C) Weak acid, strong base
- ( ) D) Weak acid, weak base
- ( ) E) Weak base, weak base
- ( ) F) Strong acid, weak acid

Use the image to answer questions 82-85



82. (1.00 pts) What is the pH at the first equivalence point?

Expected Answer: Accept anything between 3.2-3.8

83. (1.00 pts) What is the pH of the second equivalence point?

Expected Answer: Accept anything between 7.5-8

84. (2.00 pts) Explain why there are two bumps in this titration graph instead of one.

Expected Answer: The acid is a diprotic acid (1), meaning it has two hydrogen ions that can disassociate from it (1).

85. (1.00 pts) Based on the graph, a _____is being titrated with a _____ ?

- A) Strong acid, strong base
- B) Weak acid, weak base
- C) Strong acid, weak base
- D) Weak acid, strong base
- E) Weak base, strong base
- F) Strong base, weak base

86. (1.00 pts) At what temperature, in Kelvin, is water most dense? Round to the nearest whole number and do not include units in your answer.

277

87. (1.00 pts) Under normal conditions, aragonite saturation will not exceed _____Ω in a solution of water.

1

88. (1.00 pts) An influx of fecal coliform into the Mississippi River is likely a _____ source of pollution

- A) point
- B) nonpoint

89. (1.00 pts) What are the units for pH?

- A) mg/L
- B) M
- C) mM
- D) moles
- E) g/L
- F) It is a unitless quantity

90. (1.00 pts) What location is commonly used to test the levels of dissolved carbon dioxide (and therefore pH) in the ocean by most international organizations?

- A) Nassau, Bahamas
- B) Jakarta, Indonesia
- C) Dublin, Ireland
- D) Sydney, Australia
- E) Mauna Loa, Hawaii, United States
- F) Cape Town, South Africa

91. (1.00 pts) A sample of salt water has a pH of 7.4. What is its pOH? (Mind your significant figures for this and following questions!)

6.6

92. (2.00 pts) A sample of salt water has a [H+] of $5.26*10^{-8}$. What is the [H_30+]? Ensure sig figs are accurately recorded.

Expected Answer: 5.26*10^-8

93. (2.00 pts) A sample of salt water has a [H+] of $5.26*10^{-8}$. What is the [OH-]? Ensure sig figs are accurately recorded.

Expected Answer: 1.90*10-7

94. (2.00 pts) A sample of salt water has a [H+] of $5.26*10^{-8}$. What is the pOH to two decimal places?

6.72

95. (3.00 pts)
In one of the 'deserts of the ocean,' would you expect high or low light penetration, organic N, and dissolved oxygen? Enter 'High' or 'Low' into the respective blanks for each analyte

High | Low | Low

96. (2.00 pts) Homiohaline solutions have a _____ salinity while polyhaline solutions have a _____ salinity (2)

Expected Answer: Constant(1), changing(1) (other synonyms may be acceptable)

97. (2.00 pts) _____ is the condition where dissolved oxygen concentrations in water fall below 30% saturation, which is typically below 2 mg/L

- A) Hyperoxia
- B) Meromictic
- C) Anoxia
- D) Hypoxia
- E) Mortichnia
- F) Oxocline

98. (9.00 pts) Describe the zone of hypoxia in the Gulf of Mexico. What causes this zone of hypoxia? Is it seasonal or year-round? Why might the label 'dead zone' be misleading?

Expected Answer: The Gulf of Mexico is fed by the Mississippi River (1). Nutrients from agriculture in the region runoff into the river which then enters the Gulf of Mexico (1). Algal blooms consume these nutrients (1) and lower the dissolved oxygen levels in the area (1). The dead zone is seasonal (1) with its peak occurring during summer (1) due to farming practices and weather events (1). It is not a 'dead zone' because many animals are able to survive in the environment (1), but many fish cannot (1)

99. (1.00 pts) _____ _____ is the term used to describe when thermal pollution increases the overall health of an ecosystem

Thermal | enrichment

100. (1.00 pts) Hydroxyapatite is another name for _____ _____

calcium | phosphate

101. (1.00 pts) What is the current pH of the ocean?

- A) 8.1
- B) 7.5
- C) 7.0
- D) 6.7
- E) 6.3
- F) 5.8

102. (1.00 pts) Salinity influences all of the following properties of water EXCEPT

- A) Heat Capacity
- B) Electrical conductivity
- C) Density
- D) Dissolved Oxygen
- E) All of the above
- F) None of the above

103. (1.00 pts) Which of the following is NOT a way in which sedimentation pollution can harm fish populations?

- A) Clogging fish gills
- B) Reducing resistance to diesease
- C) Lowering growth rates
- D) Affecting fish egg and larvae development
- E) Impairing digestive system function

104. (1.00 pts) Dissolved oxygen concentrations can be decreased by all of the following EXCEPT

- A) Increased light intensity
- B) Increased wind
- C) Decreased salinity
- D) Increased turbidity
- E) None of the above
- F) All of the above

105. (1.00 pts) High levels of fecal coliform in water are associated with all of the following EXCEPT

- A) Tuberculosis
- B) Typhoid fever
- C) Dysentry
- D) Gastroenteritis
- E) Hepatitis
- F) All of the above are associated with Typhoid Fever

106. (2.00 pts) What is the difference between a hydrometer and a salinometer?

Expected Answer: A hydrometer measures specific gravity (1) while a salinometer measures salinity (1)

107. (1.00 pts) Select each option that is used to test for phosphates, if any

(Mark **ALL** correct answers)

- [ ] A) Jackson Candle Method
- [x] B) Vanadomolybdophosphoric Acid Method
- [x] C) Stannous Chloride Method

- [ ] D) Multiple Tube Dilution
- [ ] E) Phenoldisulphonic Acid (PDA) Method
- [ ] F) Ultraviolet Spectrophotometric Method

108. (1.00 pts) How many times more acidic is a solution of pH 4 than a solution of pH 8? If the answer is 1000 or greater, enter it without a comma (e.g. '1000' instead of '1,000')

10000

109. (3.00 pts) Describe the steps of the Jackson Candle Method

Expected Answer: A candle is shined through a column of water (1) and the length of water needed to completely obscure the light source is measured (1). The longer the column, the lower the turbidity of the sample (1).

110. (1.00 pts)
Pipes are often color-coded to indicate what substance is traveling through them. Select every option that currently matches the pipe color with the substance that passes through it

(Mark **ALL** correct answers)
- [x] A) Silver: Steam
- [x] B) Green: Seawater
- [ ] C) Brown: Non-fuel oils
- [ ] D) Yellow: Non-flammable gases
- [ ] E) Red: Chlorine Gas
- [ ] F) Light grey: Greywater

111. (1.00 pts)
The Federal Water Pollution Control Act was amended in 1972 and is now known as the _____ _____ _____. This has made it unlawful to discharge any pollutant from a point source into navigable waters, unless a permit is obtained

Clean | Water | Act

112. (1.00 pts) Write the chemical formula for orthophosphate

Expected Answer: (PO4)3-

113. (1.00 pts) What is the EPA-approved method to measure orthophosphate?

Expected Answer: Ascorbic acid method

114. (1.00 pts) Select the options where the terms are correctly matched with their definitions

(Mark **ALL** correct answers)

- [ ] A) Seasonal thermocline: A thermocline that develops in deeper water during the fall months due to the change from warmer to colder weather
- [x] B) Nephelometric Turbidity Units: A way to measure turbidity showing the tendency of particles to scatter a light beam that is focused on them
- [ ] C) Halocline: the layer in a stratified water column in which water density increases rapidly with depth
- [x] D) Nitrification: The conversion of ammonium into nitrites
- [ ] E) Aragonite: Measurement of the levels of hydroxyapatite in a body of water
- [x] F) Brine: A solution of water with over 50ppm of salts dissolved within it

115. (1.00 pts) When seawater undergoes evaporation: Which is true?

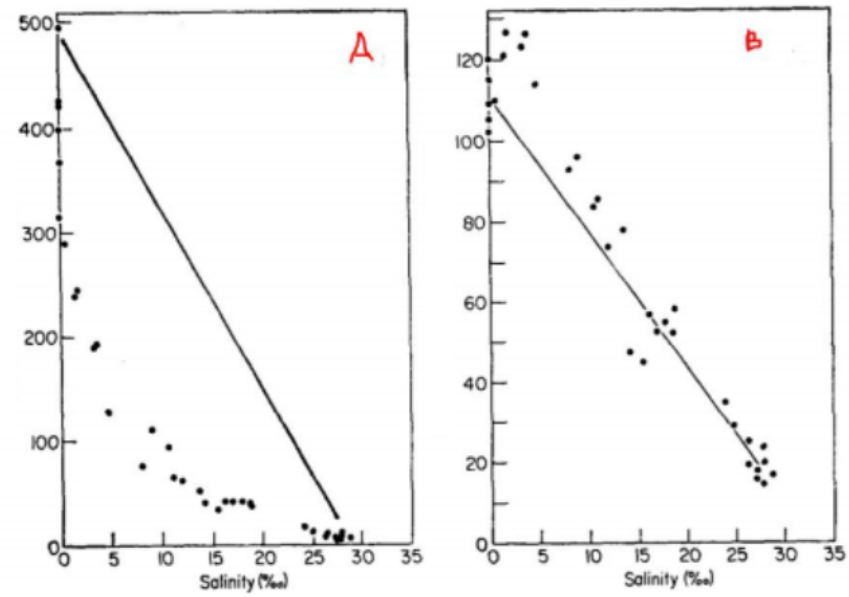
(Mark **ALL** correct answers)

- [x] A) The remaining water becomes more saline
- [ ] B) The proportions of Na and other ions change
- [x] C) The proportion of the major ions remain the same
- [ ] D) The proportion of K to other ions remain constant

116. (1.00 pts) What is the chemical formula for ammonia?

- ( ) A) NH3
- ( ) B) H2N
- ( ) C) HCN
- (•) D) NH4
- ( ) E) NH
- ( ) F) N2H4

Use this image for questions 117-118. The y-axis details the concentration of ions in mg/L



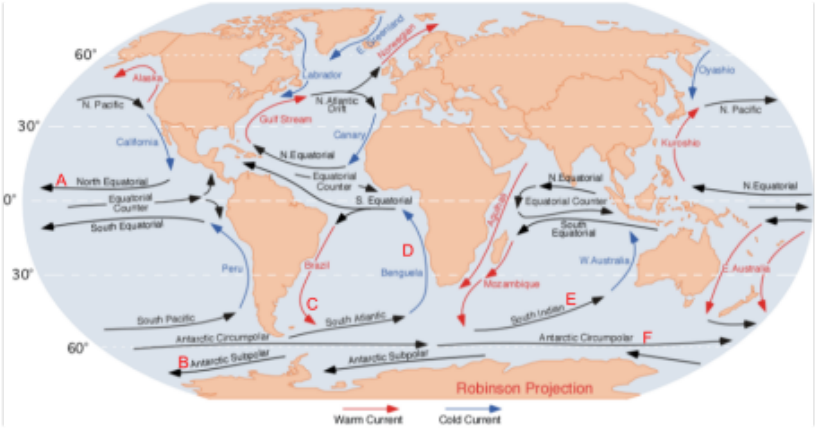
117. (1.00 pts) What ion would be represented by graph A? Please state the ion's full name, not the chemical symbol.

Iron

118. (1.00 pts) What ion would be represented by graph B? Please state the ion's full name, not the chemical symbol.

Manganese

Use this image for questions 119-120



119. (1.00 pts) Which region labeled with a letter is located in a zone of nutrient upwelling?

- A) A
- B) B
- C) C
- D) D
- E) E
- F) F

120. (1.00 pts) Which region labeled with a letter would you expect the lowest productivity?

- A) A
- B) B
- C) C
- D) D
- E) E
- F) F

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